DOD-M-82730(OS)
17 AUGUST 1984
Superseding
WS 14412, dated
27 April 1973

MILITARY SPECIFICATION

4,4-METHYLENEBIS(2,6-DI-TERT-BUTYLPHENOL)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.2 Scope. This specification covers the minimum requirements for the procurement of a low volatile, temperature stable antioxidant, 4,4'-methylenebis(2,6-di-tert-butylphenol) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1218	ACS Chemicals

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

FSC 6810

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 1744 Water in Liquid Petroleum Products by Karl Fischer Reagent

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Composition and purity. The 4,4'-methylenebis(2,6-di-tert-butylphenol) shall be a dry crystalline solid at room temperature and shall be of 98 percent minimum purity (see 4.6.1).

3.3 Melting point. The melting point shall be not less than 154.0° Celsius (C) nor greater than 155.2°C (see 4.6.2).

3.4 Benzene insolubles. The benzene insolubles shall be not greater than 0.2 percent (see 4.6.3).

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3.5 Water content. The water content shall be a maximum of 0.05% (see 4.6.4).

3.6 Workmanship. The 4,4'-methylenebis(2,6-di-tert-butylphenol) shall be uniform in quality and free from materials that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), inspection conditions shall be as specified in the applicable test method.

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on material which has been produced from raw materials with equipment and procedures normally used in production. When specified in the contract (see 6.2), the contractor shall prepare a report giving the results obtained for all inspections and tests performed. Unless otherwise specified all chemicals shall be ACS grade in accordance with MIL-STD-1218 and distilled water shall be used. Where applicable, blank determinations shall be run and corrections applied where significant. First article approval is valid on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Sample size. Sample size shall be as specified in the contract (see 6.2).

4.4.2 Inspection routine. The sample shall be subject to the tests of 4.6 plus any additional tests specified by the procuring activity (see 6.2).

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4.4.3 Failure. Failure of the sample to pass any test shall be cause for first article rejection.

4.5 Quality conformance inspections. The material shall be subjected to all the following inspections and tests for acceptance. When specified in the contract (see 6.2), the contractor shall prepare a report giving the results obtained for all inspections and tests performed and a certified statement that the lot meets all the requirements of this specification. Unless otherwise specified, all chemicals shall be ACS grade in accordance with MIL-STD-1218 and distilled water shall be used. Where applicable, blank determinations shall be run and corrections applied where significant.

4.5.1 Sampling.

4.5.1.1 Lot. For the purpose of sampling, a lot shall consist of material of a single form manufactured under one set of conditions from one production batch and offered for delivery at one time.

4.5.1.2 Sampling for examination of product. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105, inspection level II.

4.5.1.3 Sampling for tests. A representative sample of sufficient material to perform the required tests shall be taken from each lot.

4.6 Test methods. Tests shall be performed in accordance with the following methods.

4.6.1 Purity

4.6.1.1 Apparatus.

a. Beckman DK2 recording spectrophotometer or equivalent
b. Absolute ethyl alcohol
c. The usual laboratory equipment for weighing and diluting the samples.

4.6.1.2 Procedures

a. Weigh 400 ± 1 milligrams of the antioxidant sample into a 500-milliliter (ml) volumetric flask and fill to the mark with ethyl alcohol. Allow the sample to dissolve in the alcohol.

b. Dilute the sample 1:10 by pipeting 25 ml into a 250-ml volumetric flask and diluting to volume with alcohol.

c. Prepare a series of standards from the sample from step "b" as follows:

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5 ml aliquot to 50 ml with alcohol
10 ml aliquot to 50 ml with alcohol
15 ml aliquot to 50 ml with alcohol
20 ml aliquot to 50 ml with alcohol
30 ml aliquot to 50 ml with alcohol
40 ml aliquot to 50 ml with alcohol
sample with no dilution.

d. Place ethanol in matched 1-centimeter cells and set the instrument from 0 percent transmittance at 277.5 millimicrons (mu) in accordance with the manufacturers instructions.

e. Scan each of the samples from step "c", and determine its absorbance at 277.5 mu by formula

$$A = \log \frac{100}{T}$$

Where:

A = absorbance
T = percent transmittance

f. Plot the absorbance values from "e" as ordinates and concentrations from "c" as abscissa on graph paper and draw the best straight line through the points using the origin as one point.

g. Read the absorbance at the highest concentration. This is obtained from the point of intersection of the line at the most concentrated sample.

h. Calculate as follows:

$$\% \text{ purity} = \frac{A}{3567 \times 1.88 \times 10^{-6}}$$

Where:

A = absorbance from step "g"

3567 = molar absorbance index determined from a pure sample of 4,4"-methylenebis(2,6-di-tert-butylphenol)

1.88 = concentration factor in moles per liter

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4.6.2 Melting point.

4.6.2.1 Apparatus.

a. A certified thermometer graduated in 0.2° in the 150° to 155° range.

b. A second short thermometer to monitor the oil bath temperature.

c. An oil bath. (A 250-ml metal or wrapped glass beaker with mineral oil and an overhead stirrer is satisfactory.)

d. A heater in series with a variac to control heat input to the bath.

e. Capillary melting point tubes.

4.6.2.2 Procedure.

a. Place the bath on the heater and with the short thermometer in the bath and with continous stirring stabilize the temperature at near 150°C so that there is no rapid rise or fall in temperature over a 2 or 3 minute period.

b. Place antioxidant in a capillary melting point tube to a depth of about 5 millimeters and fasten the tube to the thermometer with a collar cut from a piece of rubber tubing of appropriate size.

c. Lower the thermometer into the bath at about 150°C and slowly increase the heat to give a temperature rise of about 1° per minute.

d. Read the temperature at which the sample is completely melted. This should be not less than 154°C nor greater than 155.2°C.

e. If the melting point is not within the required range, repeat steps "a" through "d" until a melting point has been determined with certainty.

Make a stem temperature correction if the thermometer used has not been calibrated for the emergent stem. The melting point shall be the average of two determinations.

4.6.3 Benzene insoluble material. Take a sample of material whose weight is between 9.500g and 10.500g. Dissolve the sample in 102.5ml of benzene. Filter the mixture through a fluted filter paper which has been dried to constant weight in a steam heated oven. Wash the residue with 50 ml of benzene. Dry the fluted filter paper and residue to constant weight in a steam heated oven. Calculate the percent benzene insoluble material by the following formula:

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$$\text{Percent} = \frac{R}{W} \times 100$$

Where: R = weight of residue in g to the nearest milligram (mg)
W = weight of sample in g to the nearest milligram

NOTE: All operations using benzene shall be performed in a hood.

4.6.4 Water content. Determine the water content of the sample by method of ASTM D-1744. Sample size shall range between 5 and 10 grams. Commercial Karl Fischer reagent with a minimum titer of 5.0 mg of water per ml may be used.

4.6.5 Examination of product. The samples taken in accordance with 4.5.1.2 shall be inspected to verify compliance with the requirements of section 5 and other requirements of this specification not involving tests.

5. PACKAGING

5.1 Packaging.

5.1.1 Commercial. Unless otherwise specified, the material shall be packaged in accordance with the manufacturer's best accepted commercial practice.

5.2 Packing.

5.2.1 Commercial. Unless otherwise specified, the material shall be packed in shipping containers in such a manner as to insure safe transportation by common carrier. Containers used shall comply with National Motor Freight Classification Rules and the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The 4,4'-methylenbis(2,6-di-tert-butylphenol) covered by this specification is intended for use in the manufacture of liner material for torpedo warheads and as a component of plastic-bonded explosive formulations.

6.2 Ordering data. Acquisition documents should specify the following:

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6.2.1 Acquisition requirements.

a. Title, number, and date of this specification
b. Quantity desired
c. Inspection conditions, if other than as specified (see 4.3)
d. First article sample size, if required (see 4.4.1)
e. Additional first article tests, if any (see 4.4.2)
f. First article instructions regarding arrangements for examinations, test and approval
g. Request for tests results, certificate, if required, (see 4.4, 4.5 and 6.2.2)
h. Any special marking required (see 5.3).

6.2.2 Contract Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of 27.410-6 of the DoD FAR Supplement are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4	Test report	DI-T-2072
4.5	Certification data/report	UDI-A-23264

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol II, AMSDL. Copies of data item descriptions required by the contractors in connections with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Preparing activity:
Navy - OS

(Project 6810-NB35)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-M-82730	4, 4-METHYLENEBIS (2, 6-DI-TERT-BUTYLPHENOL)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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